Kneeview: An Open-Source Repository of Patient-Specific Knee Geometries and Structured Meshes

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Introduction

The use of finite-element (FE) modelling to investigate knee biomechanics is often constrained by data availability. While the osteoarthritis initiative (OAI) [1] provides invaluable imaging data, converting these images into FE meshes is time-consuming. Existing open models such as Open Knee(s) [2] represent important contributions to the field but are restricted by unstructured meshes and low subject count (n=8). To address this, we present a repository of 80 patient-specific bone geometries of the femur, tibia, patella and fibula that is being expanded with personalized structured FE models through mesh morphing.

Methods

The patient specific mesh repository was built from our dataset consisting of T2-weighted, fat saturated, knee MRI scans from 93 patients (38 male, 55 female) acquired at Hospital del Mar, Barcelona. An nnUNet was trained on 35 expert annotations and evaluated on a 15-volume hold-out test set, both verified by a specialized musculoskeletal radiologist, and compared against four foundation models (FMs), TotalSegmentator (TS), MEDSAM2, MRI-Core and SegmentAnyBone. After excluding subjects with missing demographics, 80 segmentations were retained. We adapted the Bayesian Coherent Point Drift (BCPD++) algorithm for tissue-specific morphing of bones, with automatic interpolation for soft tissues. The repository is hosted on the Kneeview webservice (https://knee.view.upf.edu).

Results

The nnUNet achieved high segmentation performance on a hold-out test dataset of 15 volumes with an average Dice score of 0.986 ± 0.001, IoU of 0.971 ± 0.002 and Hausdorff Distance (HD95) of 14.464 ± 12.727 mm. This HD95 variance was driven by distal tibial artefacts, which drops to 1.74 ± 0.060 mm after their removal. Among the FMs, TS showed the best performance (Dice 0.92 ± 0.002, IoU 0.85 ± 0.003, HD95 7.38 ± 0.20 mm) but remained below the task-specific nnUNet. All segmented 3D bone geometries are publicly available in Kneeview. The template FE mesh included the complete knee joint with ligaments, bones and menisci and the adjusted BCPD++ algorithm generated structured meshes for all patients without any distorted or zero volume elements, achieving a scaled Jacobian above 0.9 and a mean surface distance (MSD) of approximately 1 mm relative to target segmentations.

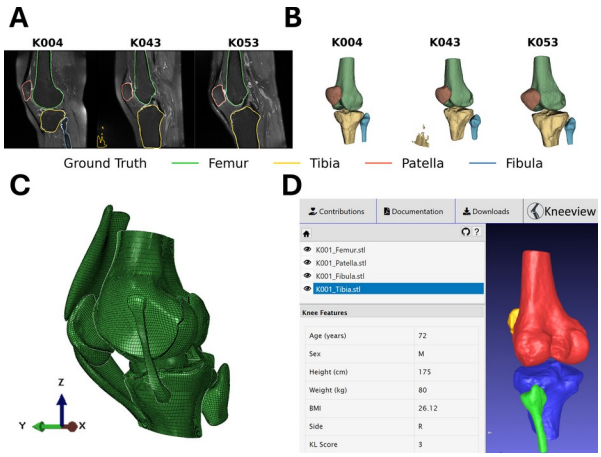


Figure 1: nnUNet segmentations and Kneeview interface. a) nnUNet segmentation versus ground truth. b) 3D visualization of segmentations. c) FE model. d) Example of Kneeview interface.

Conclusions

A collection of 80 patient-specific segmentations was successfully created and is accessible through the Kneeview interface. The high Dice scores confirm that our automated pipeline produces bone geometries comparable to manual segmentation, and structured FE meshes were generated for all subjects. As Kneeview grows to include these meshes, it aims to help address the shortage of open-source simulation-ready knee models. Future work will focus on validating the morphed models and running FE simulations of full gait cycles for all subjects before releasing simulation-ready models into the Kneeview interface.

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